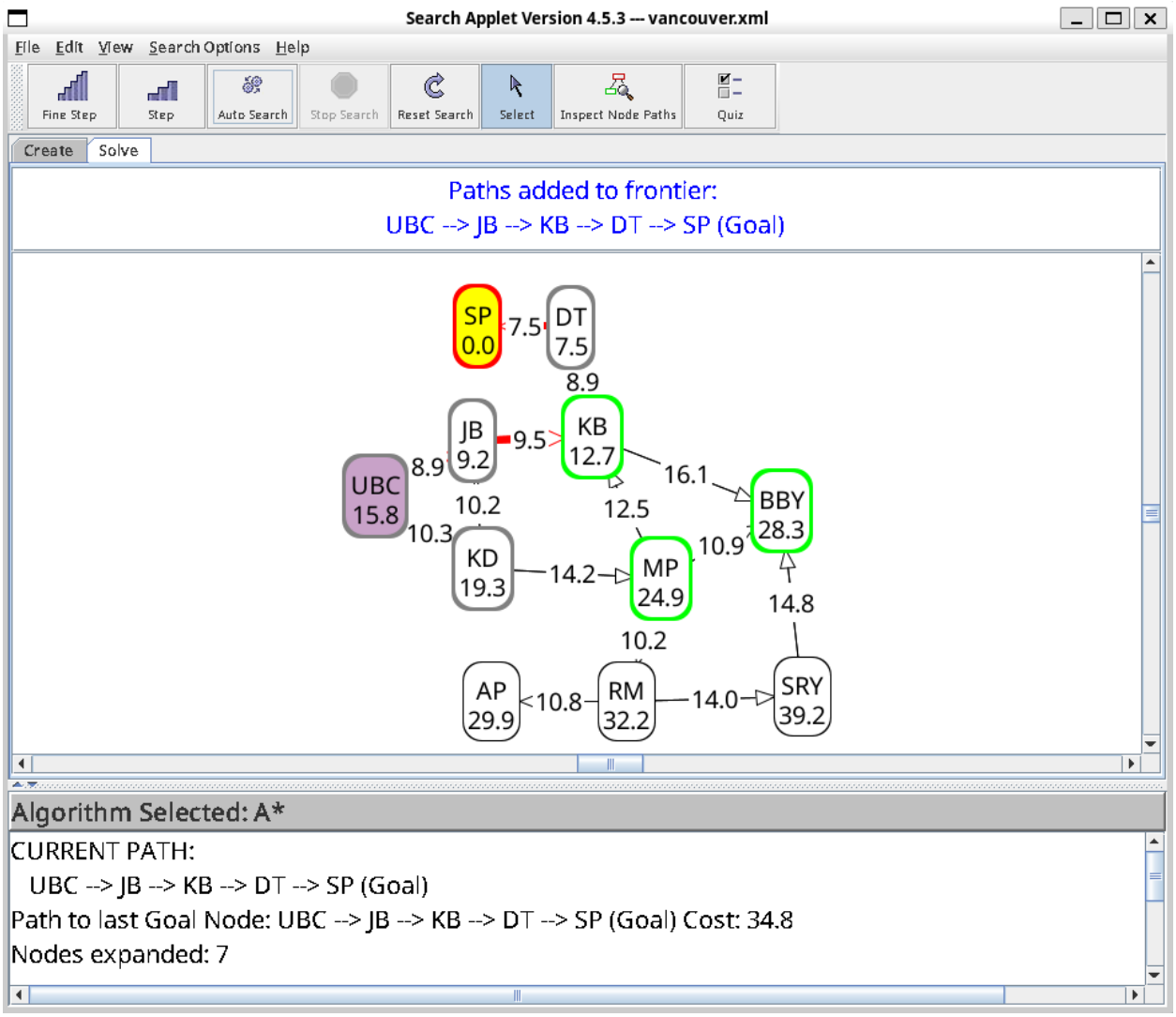
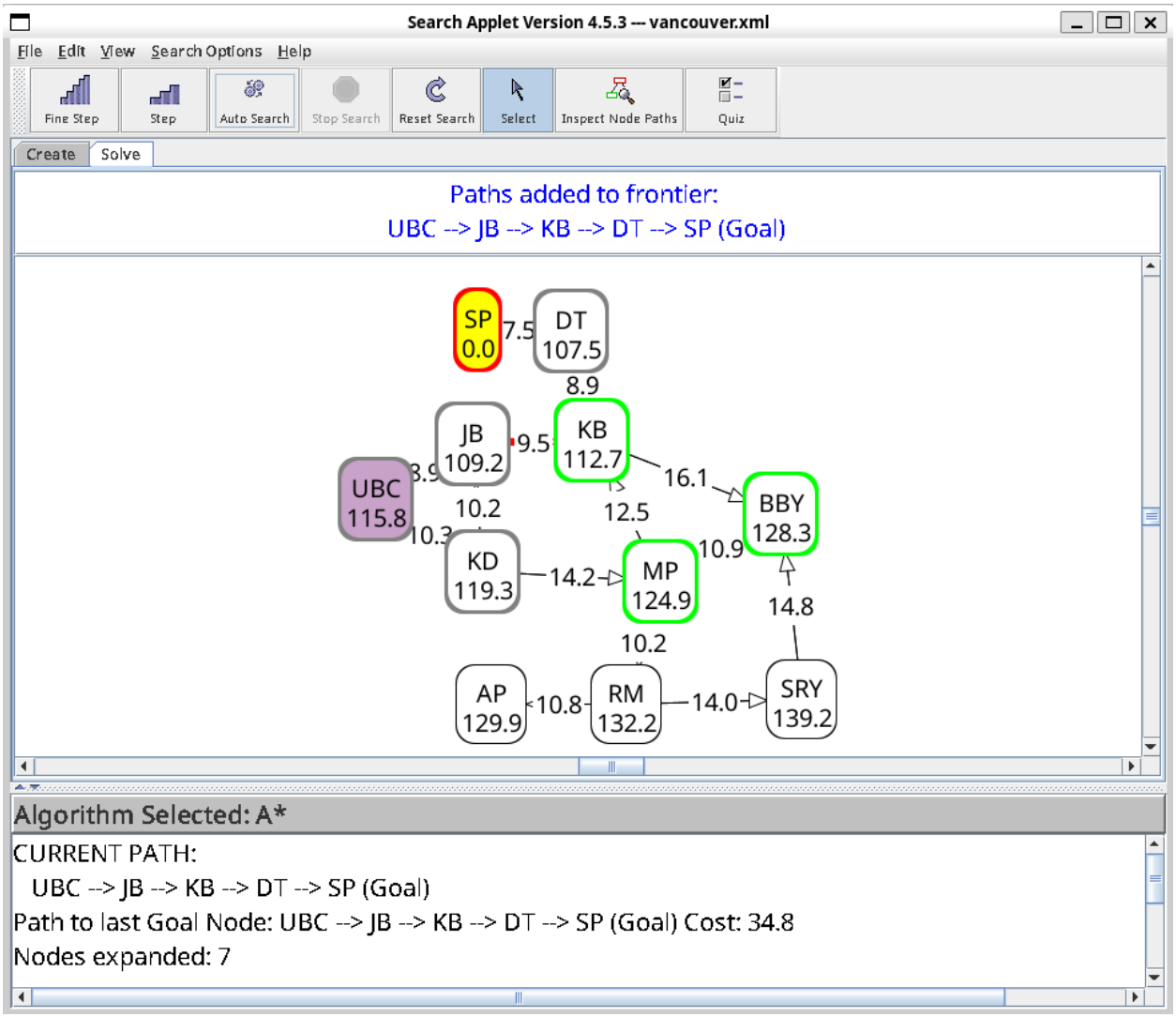
Report

Reference heuristics

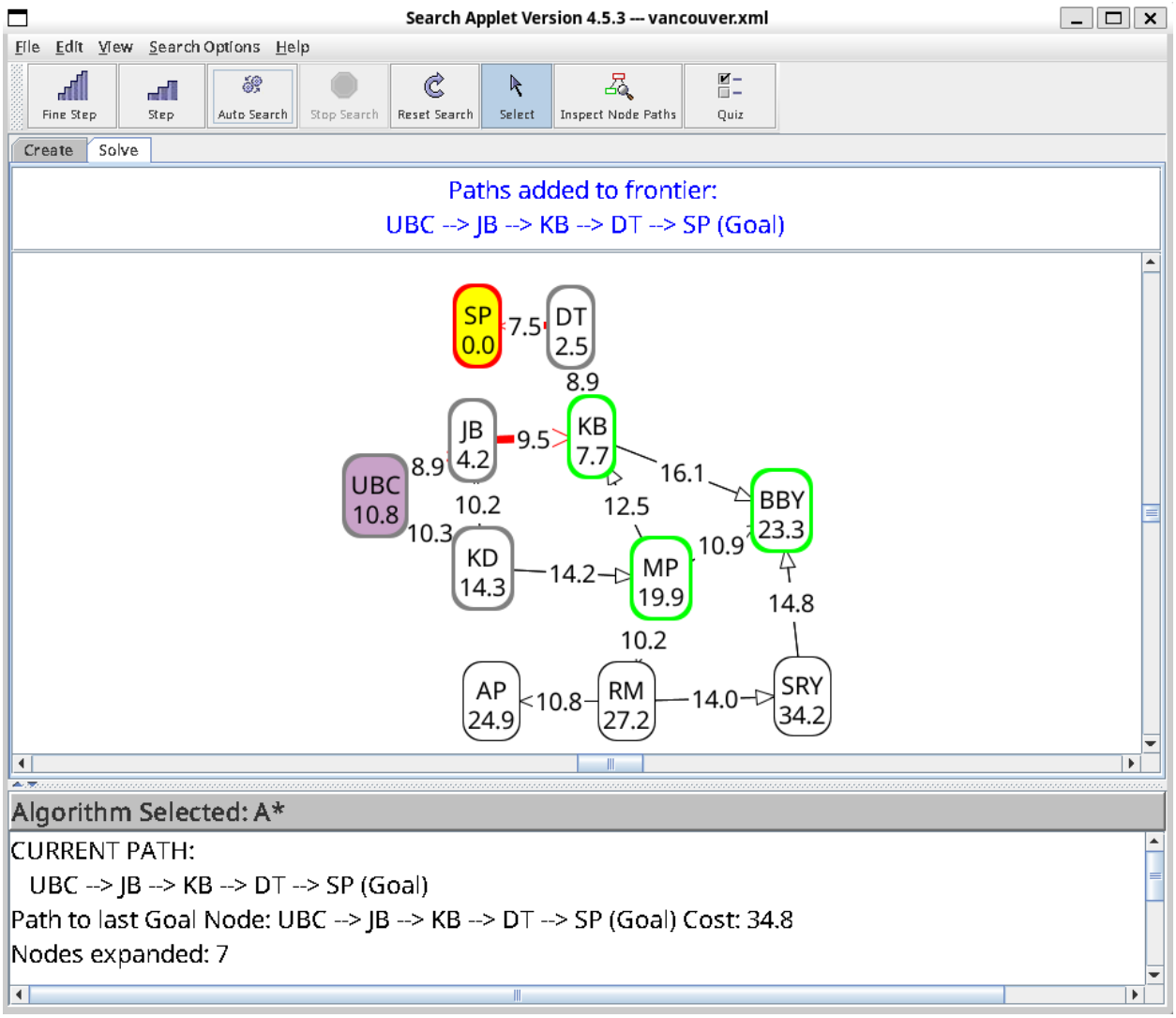


Increase heuristics everywhere (all by 100, except goal)



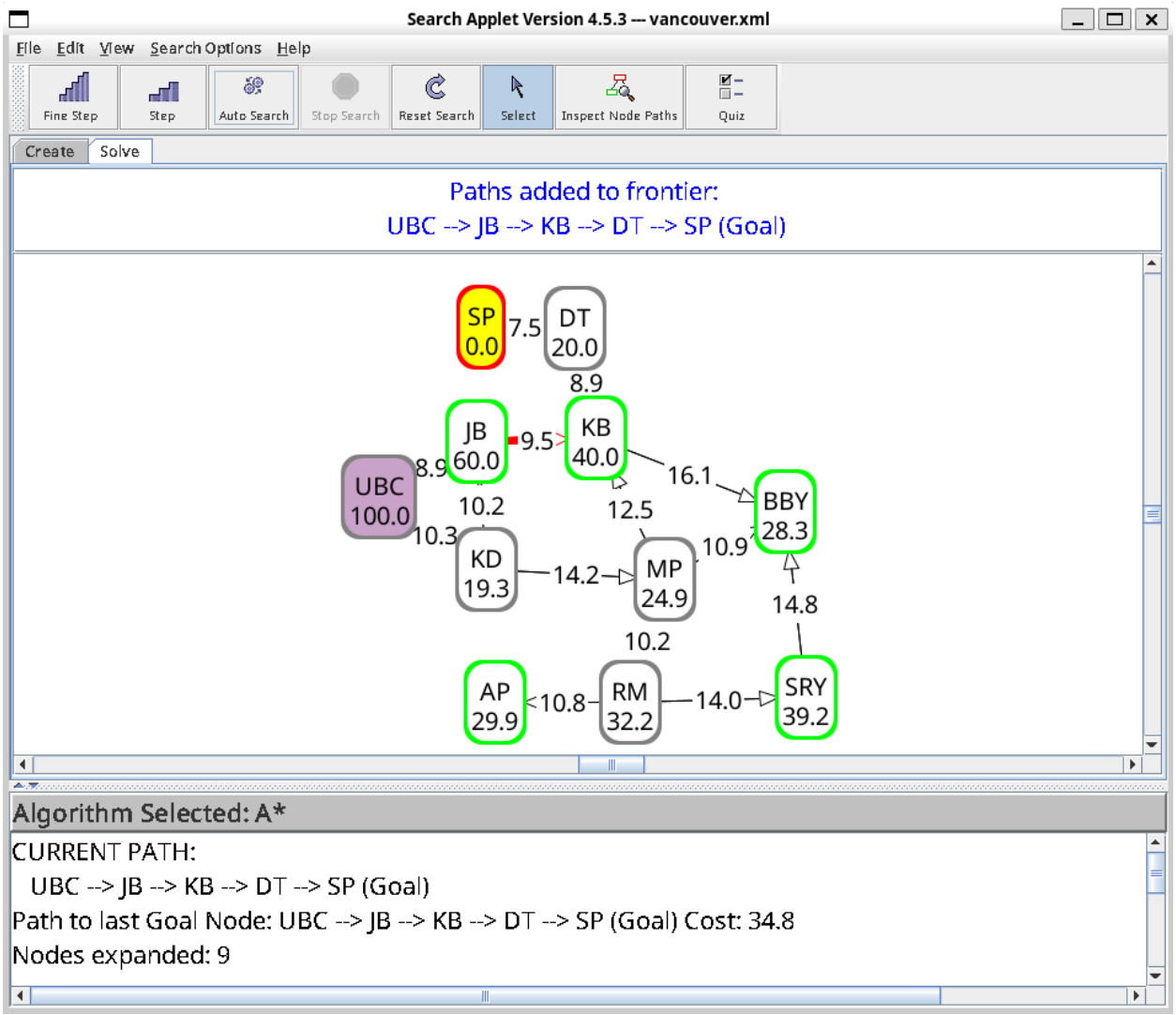
result: no change

Decrease heuristics everywhere (all by 5, except goal)



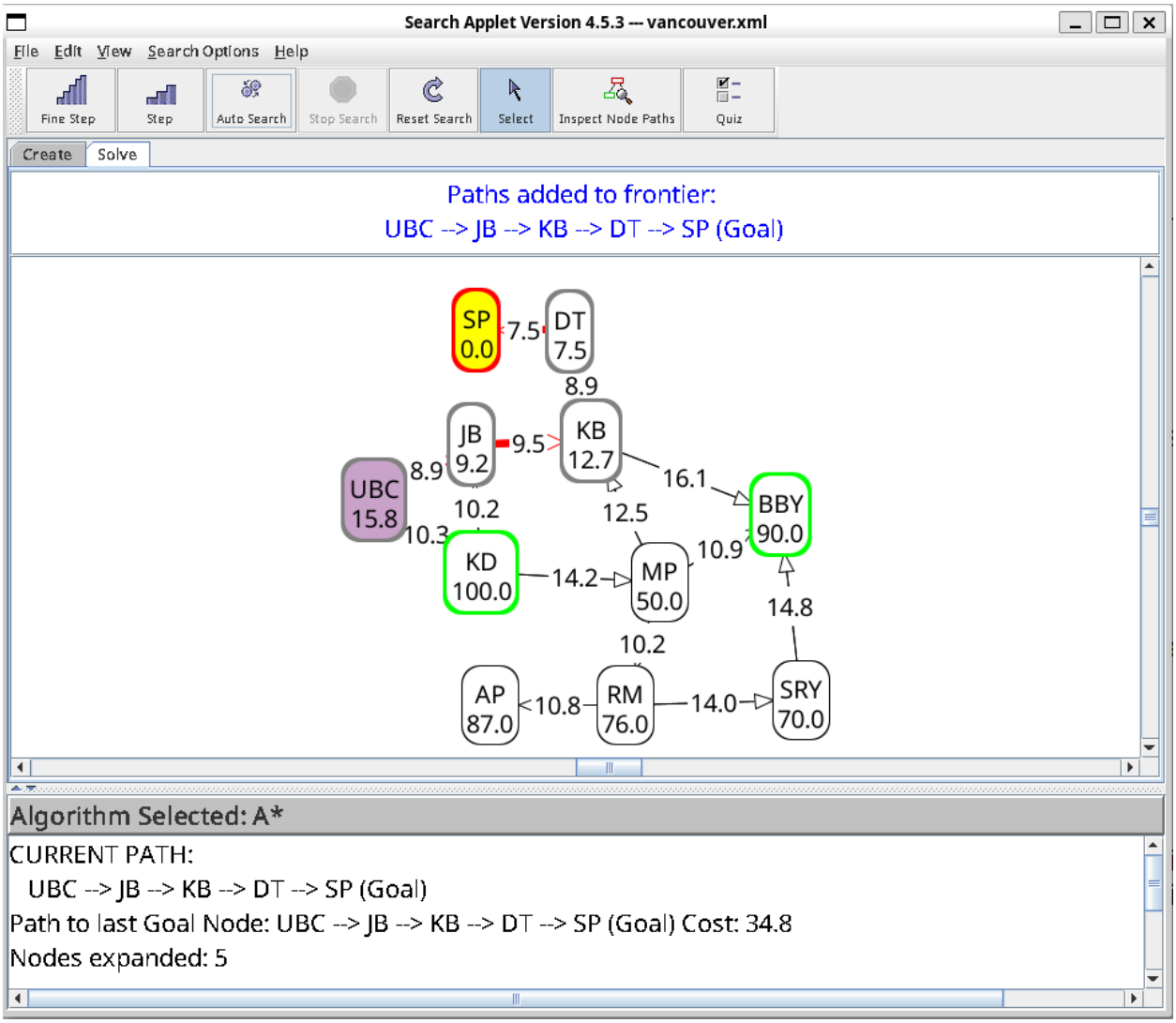
result: no change

Changing heuristics on solution path (by significantly increasing)



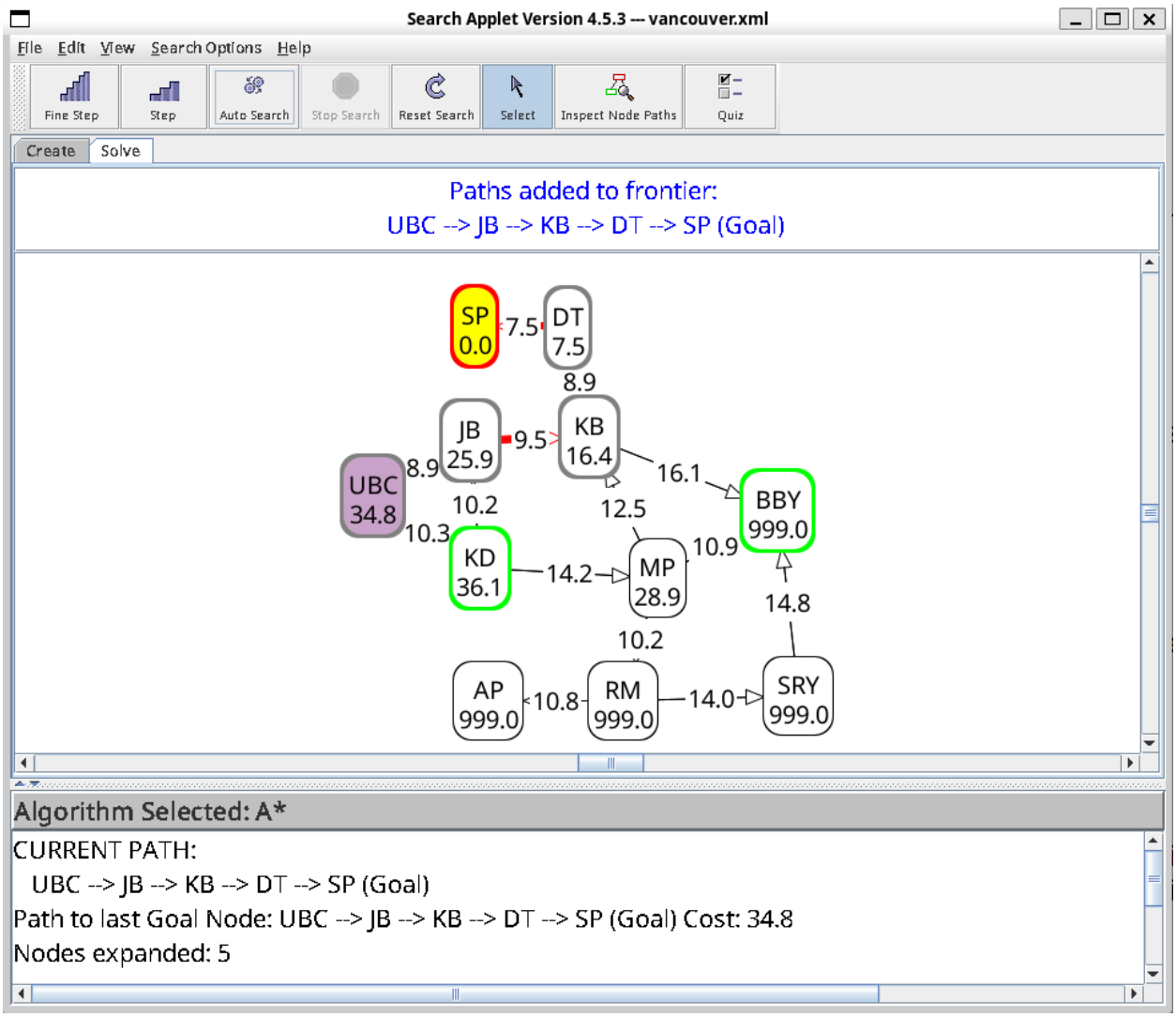
result: slightly more nodes expanded, no change to optimal path

Changing heuristics on nodes that are not on solution path (also by significantly increasing)



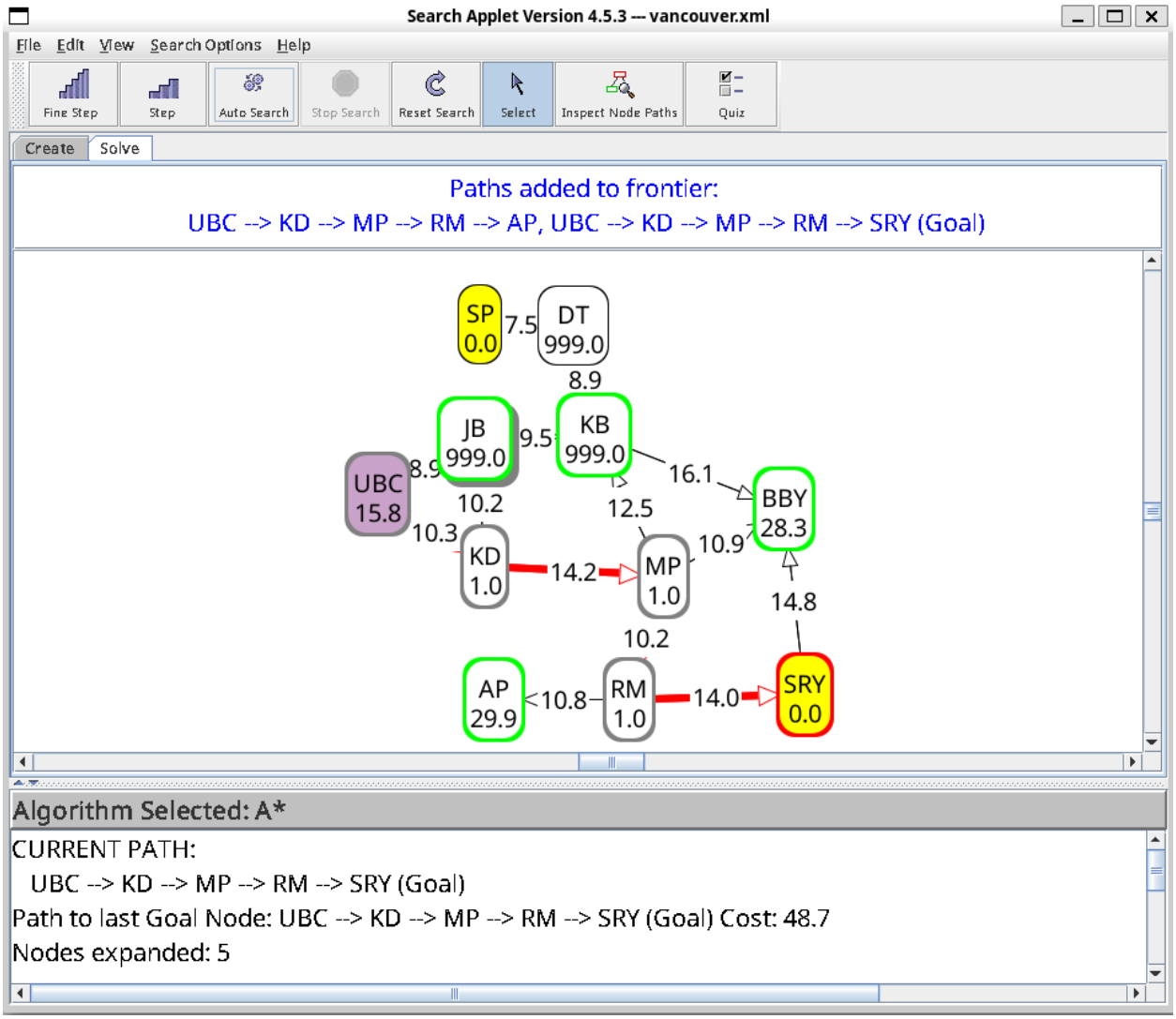
result: less nodes expanded, no change to solution path

Changing heuristics to the exact minimum costs to goal node (for nodes that are unable to reach the goal, I set them to 999)



result: less nodes expanded, no change to solution path

Making a second goal node, and modify heuristics to make the solution path change to SRY



this is achievable. If I make the heuristics on the original solution path absurdly high (999) and the ones on the path to SRY absurdly low (1), I can make the algorithm return a path to SRY instead of SP.

(Of course, we need to set the heuristic of SRY to 0, too)

Conclusion

Tighter heuristics will make the A* algorithm expand less nodes before finding the goal but only changing the heuristics won't alter the solution path. If we allow a second goal node, modifying heuristics will introduce a different solution path.